Phase 4.25: Complete Rank 1–40 Coverage and the Rank-34 BVE Frontier

Tactix development report for Andrew

30 August 2026

1 Executive outcome

The current homogeneous Rank 1–40 candidate solves every instance with the expected verdict. All 22 SAT models and all 18 UNSAT proofs have now passed independent validation, and all 40 result records report certificate and derived-artifact cleanup with no residue. Campaign 1657091 therefore closes the missing-verdict objective on this evaluation set.

The candidate takes 9,305.953 seconds, versus 9,987.021 seconds for its immediate predecessor, 11,222.888 seconds for frozen accepted Tactix, and 7,675.236 seconds for pinned Kissat. It is 0.931803x the predecessor and 0.829189x accepted Tactix, but remains 1.212466x Kissat. SAT aggregate time is 4,083.055 seconds versus Kissat's 4,057.914 seconds; the remaining aggregate gap is primarily UNSAT, where Tactix is 1.443858x Kissat by total time.

This is a coverage and correctness closure, not yet the Phase-4.25 performance exit. The candidate binary is immutable SHA-256 `1d85a694a122e3a466870b598e82159442196ffef7a5d37343912021678cb4f7`. Newer Congruence and BVE parity work is deliberately unpromoted and is not included in these aggregate numbers.

2 Major progress since the 27 August report

2.1 Uniform strategy and ordering contract

Production now uses Kissat creation order unconditionally. CaDiCaL arena order remains available only through the explicit diagnostic arm `TACTIX_ARENA_ORDER=1`. Formula-dependent production profiles and overrides remain removed. BVE, Factor, forward subsumption, and Vivify are uniformly enabled at 100, 50, 100, and 100 permille. No retained repair uses a formula, size, density, benchmark, identity, rank, classification, admission, or effort gate. This preserves N1–N4 and makes regressions evidence of a general missing mechanism rather than grounds for restoring a selector.

2.2 General mechanism closures

The most consequential retained repairs are:

- Stable Reorder ties use permanent first-import rank, and scanning stops at the 100th open clause literal, matching Kissat's compact order.
- Packed large on-the-fly strengthening now separates physical reconstruction from semantic added/removed notifications. This closes Rank 15 without changing admission or effort.
- Duplicate AND/OR gate membership, physical large gate bases, and general-definition import now follow Kissat's physical clause image.

- Compact random-decision slots use first-import order filtered to the activated live population, while queue and score initialization retain activation order. This removes Rank 20's first search divergence.
- Strengthen's local mixed stream and proofless/DRAT binary-id storage were reduced without changing clause order or the checked search trajectory, materially lowering Rank-6 memory and runtime.

Rank 15 is now close to Kissat, Rank 20 is faster than Kissat in the homogeneous run, and Rank 36 remains close. Selected final rows are:

Rank	Verdict	Tactix (s)	Kissat (s)	Ratio
15	SAT	82.195	80.788	1.02x
20	SAT	124.566	140.142	0.89x
29	SAT	314.075	297.121	1.06x
32	UNSAT	387.998	289.685	1.34x
34	SAT	616.624	344.045	1.79x
36	SAT	383.362	368.510	1.04x

2.3 Codebase revisit

The retained diff was reread against pinned Kissat and audited for strategy ownership, physical occurrence representation, ordering, and diagnostic leakage. No formula-dependent consumer was reintroduced. The revisit also confirmed that the trace diagnostics are compiled out of Release and that the CaDiCaL order is not a hidden production fallback. The detailed audit is in `benchmarks/reports/2026-08-30-codebase-revisit-and-otfs-candidate.md`.

3 Rank-34 causal progress

Rank 34 is the largest remaining absolute SAT gap. The investigation began at the first search split and moved upstream through Substitute and Sweep to a Congruence closure defect. Tactix registered only currently canonical gate images, so a residual valued gate could disappear from the rescheduling graph. Kissat registers every gate's immutable right-hand side and moves occurrence rings when representatives merge. The general repair restores Kissat's 3,778 merges and 33 total startup units, including the causal units `-1136` and `-574`. It is retained even though the first full Rank-34 solve regressed to 744.945 seconds, because it repairs a reference invariant rather than a benchmark-specific choice.

The next owner is BVE. Several general physical-lifecycle mismatches have been closed:

- syntactic gate implication indices contain only physical binaries;
- failed candidates compact inspected physical occurrence stacks;
- an absent heap item receives a score update without reinsertion;
- newly produced root units flush their connected satisfied clauses at the same point as Kissat; and
- BVE's embedded Kitten consumes the initialization seed before its first cleared environment, while Sweep continues to use its fresh seed zero.

After these repairs, every one of the first BVE schedule's 14,582 heap pops is identical to Kissat. Both solvers then create the same 249-variable second schedule in the same insertion order. Its relevance image is exact, but three selected physical occurrence counts differ:

External variable	Kissat positive/negative	Tactix positive/negative	Delta
839	5/5	4/5	one positive missing
843	5/6	4/5	one on each polarity missing
845	4/6	4/5	one negative missing

Controls 842, 14754, and 850 are exact at 5/5, 4/6, and 7/9. This is measured evidence that the remaining boundary is first-round physical clause/occurrence lifecycle, not Factor, EVSIDS arithmetic, heap comparison, or schedule admission. The exact clause identities and their first removal/addition event are the next diagnostic target.

4 Correctness and provenance

The Rank 1–40 authority is `benchmarks/runs/20260830-2710-tierd-ranks1-40-randec-import-order/`. Its 40 result records contain expected verdicts, 22 verified models, 18 verified DRAT proofs, and zero cleanup residue. The generated full per-rank runtime/RSS comparison is `comparison-live.md` in that directory.

The Rank-34 chronology is `benchmarks/reports/2026-08-30-rank34-sat-divergence.md`. The complete first-schedule pop windows are campaigns 3060, 3190, and 3200; the selected second-schedule count comparison is campaign 3210, job 1657375. These are bounded compile-time observer runs and do not alter Release strategy, proof, admission, score, or effort.

The current unpromoted worktree has focused BVE and SubSolver tests passing. Its most recent full suite had 75 of 76 tests pass; the sole failure was the comment-hygiene spelling `AND/OR BASE`, which has been corrected but has not yet received the final full-suite rerun. Thus the homogeneous candidate's correctness is fully established, while the later Rank-34 work remains explicitly pending cumulative validation.

5 Risks and decisions

- Overall performance is still 1.212466x Kissat, above the 1.10 Phase-4.25 target. UNSAT ranks 22, 31, and 38 remain roughly 2.2x/2.2x/2.1x Kissat.
- Representation and memory remain substantially higher on several large SAT instances; complete coverage does not close the memory dimension.
- The general Congruence repair and newer BVE repairs are intentionally retained through downstream regressions. Reverting them would restore an incorrect mechanism and hide the next discrepancy.
- No broad campaign will be started merely because a bounded trace moved forward. The next production gate is a full Release suite followed by one Rank-34 model-checked timing run.
- The worktree remains uncommitted. Commit and push follow a coherent, cumulatively validated increment, not an intermediate observer result.

6 Next actions

Identify the exact clauses missing from variables 839, 843, and 845 at the second BVE schedule, trace their first-round lifecycle, and repair only the general Kissat mismatch. Then run the complete Release/correctness suite and one Rank-34 Release timing/model check. If that gate is positive or clarifies a retained general repair, update the parity ledger and progress documents and decide whether one homogeneous Rank 1–40 promotion campaign is justified. Continue to prohibit formula, rank, benchmark, size, density, and classification selectors.